

RESEARCH INTEREST

- Primary Focus** Materials and Interface Engineering for Affordable and High-Performing Batteries
- Key Approaches**
- Designing battery materials through controlled disorder and surface coatings
 - Investigating chemo-mechanical challenges at solid-state battery interfaces
 - Leveraging machine learning techniques to accelerate experimental workflows

EDUCATION

Korea Advanced Institute of Science and Technology (KAIST) Mar. 2019 - Aug. 2026 (Expected)
Undergraduate Program Daejeon, South Korea

- Primary Major: **Chemical and Biomolecular Engineering**
- Double Major: **Materials Science and Engineering**
- Overall GPA: **4.21/4.3** (3.99 on 4.0 scale) — CBE GPA: 4.19/4.3, MSE GPA: 4.3/4.3
- Leave of Absence (Fall 2022 - Spring 2024) due to the mandatory military service

Georgia Institute of Technology Aug. 2024 - May 2025
Exchange Program in Chemical & Biomolecular Engineering Atlanta, Georgia, U.S.A.

- Overall GPA: **4.0/4.0**

RESEARCH EXPERIENCE

Energy Materials Lab — Prof. Nam-Soon Choi Jul. 2025 - Present
Undergraduate Research Assistant Dept. of CBE, KAIST

Project: Electrolyte Additive Exploration for Low-Temperature Na-ion Batteries

- Acquired insights into fundamental properties of electrolytes and additives for Na-ion batteries

Liu Resarch Group — Prof. Meilin Liu Aug. 2024 - May 2025
Undergraduate Research Assistant School of MSE, Georgia Tech

Project: Surface Treatment on Ni/Mn-based Co-free Layered Cathode for Li-ion Batteries

- Devised and applied Lithium-Molybdenum-Phosphate (LMP) surface treatment strategy to tackle inherent instability of Co-free chemistry, achieving doubled specific capacity during high-rate operation
- Proposed dual-mechanism of Li_3PO_4 surface layer formation and Mo-ion doping through microscopic characterization

Electrochemical Energy Devices Lab — Prof. Hee-Tak Kim Jan. 2021 - Dec. 2021
Undergraduate Research Assistant Dept. of CBE, KAIST

Project 1: Cathode Materials Optimization for Flowless Zn-Br Aqueous Batteries

- Screened various additives, including conductive carbons and catalysts, to optimize performance

Project 2: Degradation Analysis and Lifetime Enhancement of Zn-MnO₂ Aqueous Batteries

- Implemented electrolyte reservoir to mitigate premature cell failure due to H₂ evolution in flowless design

PUBLICATION

1. Y. Ahn, X. Hu, Y. Ding, C. Kim, Y.-C. Wu, **T. Kim**, M. Liu*, **Synergistic Effects of Entropy Tuning in Niobium-Based Oxide Anode for Fast-Charging Lithium-Ion Batteries.** *Adv. Funct. Mater.* 2025, e09533. <https://doi.org/10.1002/adfm.202509533>

TECHNICAL SKILL

Synthesis & Fabrication	Solid-state Synthesis, Slurry Casting, Coin Cell Assembly
Electrochemical Analysis	Galvanostatic Cycling, Cyclic Voltammetry, EIS, GITT
Materials Characterization	<i>Hands-on:</i> XRD <i>Data Analysis:</i> XPS, Raman, SEM
Software & Programming	Origin, Python (Machine Learning), MATLAB, $\text{\LaTeX}$ • Developed surrogate models to predict CO₂ capture process (Course project in <i>Data Analytics for Chemical Engineers</i>)

ACADEMIC PRESENTATION & AWARD

Poster Presentations

- 12th Annual MRS Poster Competition**, *MRS Chapter at Georgia Tech* Apr. 2025
- 19th UROP Spring Symposium**, *Undergraduate Research Opportunities, Georgia Tech*
- *Title:* Effect of LMP Surface Coatings on Co-Free Layered Cathode for Li-ion Batteries

Academic Awards

- Best Study Award**, *KPF Student Academic Conference, KAIST* Feb. 2022
- Third Prize out of 26 teams**
- *Title:* Developing Electrolytic Zn-Mn Battery System and Identifying Key Operation Conditions

HONOR & SCHOLARSHIP

- KAIST Presidential Fellowship (KPF)**, *KAIST* Mar. 2021 - Present
- National Presidential Science Scholarship**, *Korea Student Aid Foundation* Mar. 2021 - Dec. 2024
- KOR-US STEM Student Exchange Scholarship**, *Ministry of Trade and Industry* Apr. 2024
- Merit-Based Academic Excellence Scholarship**, *Dept. of CBE, KAIST*
- *Awarded semesters:* Fall 2020 (2nd place), Spring 2021 (1st place), Fall 2021 (1st place)
- Dean's List**, *College of Engineering, KAIST*
- *Awarded semesters:* Fall 2019, Spring 2020, Fall 2020, Spring 2021, Fall 2021, Spring 2022

EXTRACURRICULAR ACTIVITY

- International House Resident**, *Georgia Tech* Jan. 2025 - May. 2025
- Engaged with diverse cultures and strengthened global communication skills
- Social Service Personnel**, *Republic of Korea Army* Nov. 2022 - Aug. 2024
- Alternative mandatory military service; Served at local social welfare center
- Calculus II Tutor for Freshmen**, *KAIST* Mar. 2020 - Jun. 2022
- Delivered short lectures and provided tutoring for 5 consecutive semesters
- Departmental Student Council**, *Dept. of CBE, KAIST* Sep. 2020 - Dec. 2021
- Served in the Division of Culture & Event
- Online IT Volunteer**, *Worlds Friends Korea* Nov. 2020
- Designed and delivered lectures about Adobe Photoshop skills to Ugandan